# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE CODE: MLA03**
**COURSE NAME: Reinforcement learning for Environment and Agent**

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**COURSE OUTCOME COVERED**

***CO2:*** *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

*Assessment Tool Weightage: 35%*

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Dr DUMPALA PRASANTH

**QUESTIONS: 10** **Total Marks: 50**

Annexure A

**Industry Problem Task** **Duration: 2.5hrs**

1. Explain how Reinforcement Learning can be applied to optimize delivery routes in a logistics system. Identify the possible states, actions, rewards, and policy. Discuss how continuous learning improves efficiency over time. **(5 Marks)**

2. Apply Reinforcement Learning concepts to a fraud detection system in fintech. Define the state space, action space, and reward mechanism. Explain how exploration strategies improve detection accuracy. **(5 Marks)**

3. Illustrate how Reinforcement Learning is used in a streaming platform for content recommendation. Discuss the impact of delayed rewards, user feedback, and changing preferences on learning performance. **(5 Marks)**

4. Evaluate the effectiveness of Reinforcement Learning in predictive maintenance compared to traditional rule-based approaches. Discuss risks and reliability concerns in industrial environments. **(5 Marks)**

5. Analyze how Reinforcement Learning can be used to optimize transportation systems such as bus scheduling or route allocation. Define states, actions, and rewards, and discuss how real-time data improves performance. **(5 Marks)**

6. Apply Reinforcement Learning to dynamic pricing in e-commerce. Define the state space, action space, and reward function. Discuss challenges in balancing exploration and exploitation. **(5 Marks)**

7. Illustrate the use of Reinforcement Learning in smart city waste management systems. Identify the MDP components and discuss how continuous updates improve operational efficiency. **(5 Marks)**

8. Evaluate how Reinforcement Learning can be used in network bandwidth allocation in telecommunications. Define states, actions, and rewards, and examine how the system adapts to dynamic conditions. **(5 Marks)**

9. Analyze the application of Reinforcement Learning in portfolio management in finance. Discuss advantages over traditional models and challenges such as risk and delayed rewards. **(5 Marks)**

10. Justify the use of Reinforcement Learning in renewable energy management systems. Define the RL framework components and discuss the impact of environmental uncertainty on decision-making. **(5 Marks)**

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***Code of Conduct:***

*I **Charandeep.n.c/1924721961** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:*** **charandeep**

## RUBRICS FOR CALCULATION

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
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| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
| Understanding of Industry Problem | 20% | Comprehensive understanding of real-world problem and constraints | Good understanding with minor gaps in requirements | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |
| Application of Engineering Concepts | 25% | Applies advanced and relevant concepts effectively to solve the problem | Applies appropriate concepts with minor errors | Limited application of concepts | Incorrect or no application of concepts |
| Solution Design | 25% | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps | Basic solution with limited feasibility | Unclear or impractical solution |
| Use of Tools | 15% | Effective use of modern tools, technologies, and real data | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools | Minimal or incorrect use of tools |
| Analysis | 10% | Thorough analysis with validated results and performance evaluation | Good analysis with reasonable validation | Basic analysis with limited validation | Poor or no analysis |
| Professionalism | 5% | Highly professional, well-documented, and clearly presented work | Good documentation and presentation | Basic documentation with some gaps | Poor documentation and presentation |

## Question 1

***Explain how Reinforcement Learning can be applied to optimize delivery routes in a logistics system. Identify the possible states, actions, rewards, and policy. Discuss how continuous learning improves efficiency over time.***

In a logistics system, Reinforcement Learning (RL) treats the delivery vehicle or fleet-management system as an agent that learns to select the most efficient sequence of routes through repeated interaction with a dynamic environment consisting of traffic, weather, delivery windows, and customer locations. Instead of relying on a fixed, pre-computed route generated once by a static shortest-path algorithm, the RL agent continuously updates its routing policy based on real-time feedback, allowing it to adapt to congestion, road closures, accidents, and changing delivery priorities as they occur throughout the day. This makes the routing decision a sequential decision-making problem rather than a one-time optimization, which is precisely the type of problem RL and dynamic programming are designed to solve.

## State Space

The state can be represented by the vehicle's current location (GPS coordinates or node in the road network), the set of remaining delivery points and their time-window constraints, the time of day, current traffic density on nearby road segments, the vehicle's fuel or battery level, cargo load, and prevailing weather conditions. Each state captures the full situational context the agent must reason about before choosing the next action, and the size of this state space grows quickly as more delivery points, vehicles, and environmental variables are added, which is why function approximation (such as deep neural networks in Deep RL) is often used instead of a simple lookup table.

## Action Space

Actions correspond to the next road segment or delivery point to travel to — for example, choosing which intersection to turn at, whether to reroute around a reported traffic jam, whether to wait at a location for a time-sensitive delivery window to open, or deciding the order in which to visit the remaining customers on the route. At a higher level of abstraction, actions may also include assigning a particular parcel to a particular vehicle in a multi-vehicle fleet.

## Reward Function

Rewards are typically designed to be positive for on-time deliveries, minimized total travel distance, and reduced fuel or energy consumption, while penalties are applied for delays, missed delivery windows, excessive idling, or unnecessary detours. Reward shaping — carefully balancing these competing objectives — directs the agent toward routes that jointly optimize speed, operating cost, and customer-service reliability rather than optimizing a single metric in isolation.

## Policy

The policy $\pi(s)$ maps the current state to the best next action, essentially deciding the next stop or road segment that maximizes expected long-term (discounted) cumulative reward rather than only the immediate distance saved on the next hop. A good policy therefore looks ahead: it may accept a slightly longer next segment now if doing so avoids a much larger delay later in the route, which is the essential advantage RL has over greedy, myopic routing heuristics.

## Continuous Learning and Efficiency

As the agent experiences thousands of deliveries across varying traffic and weather patterns, it refines its policy through trial and error (and, in practice, through offline simulation followed by online fine-tuning), gradually learning which routing decisions consistently lead to faster and cheaper deliveries. Over time this continuous learning reduces fuel costs, shortens delivery times, improves customer satisfaction and on-time delivery rates, and allows the system to automatically adapt to seasonal demand spikes, new road infrastructure, or newly opened delivery hubs without manual reprogramming of the routing rules. Companies such as large parcel-delivery and ride-hailing platforms use this type of adaptive routing to continually shrink average delivery time and cost per package as the system accumulates more operational experience.

## Question 2

***Apply Reinforcement Learning concepts to a fraud detection system in fintech. Define the state space, action space, and reward mechanism. Explain how exploration strategies improve detection accuracy.***

In fintech fraud detection, RL frames the detection engine as an agent that observes each incoming transaction (or a sequence of transactions belonging to an account) and decides whether to flag it as fraudulent, allow it, or send it for additional verification. Unlike static rule-based systems, which apply fixed thresholds such as 'block any transaction above ₹1,00,000 from a new device', the RL agent adapts as fraud patterns evolve, learning from the outcomes of its own past decisions and from confirmed fraud/legitimate labels that arrive after the fact.

## State Space

The state includes transaction attributes such as transaction amount, merchant category, transaction location and IP geolocation, device fingerprint, time since the last transaction on the account, historical spending pattern of the account holder, and behavioural features such as typing speed or navigation pattern within the banking app. Together these features describe the 'context' the agent must classify.

## Action Space

Actions include approving the transaction, blocking it outright, or routing it for manual review or step-up authentication such as an OTP or biometric challenge. Some systems also include a 'soft decline with retry' action that temporarily holds a transaction while requesting more information from the user.

## Reward Mechanism

The agent receives a positive reward for correctly approving legitimate transactions and correctly blocking fraudulent ones, and a strong negative reward for false negatives — fraud that goes undetected, causing direct financial loss and regulatory exposure — and a smaller but still meaningful negative reward for false positives, i.e., legitimate transactions that are incorrectly blocked, which harm customer experience and can lead to customer churn. This asymmetric reward structure reflects the fact that missed fraud is usually far more costly to the institution than an inconvenienced customer, so the agent is trained to be appropriately risk-averse.

## Exploration Strategies and Detection Accuracy

Because fraud patterns constantly evolve as criminals adapt to existing controls (a phenomenon known as concept drift), the agent must balance exploitation of known fraud signatures with exploration of new or borderline transaction patterns that do not yet fit any known rule. Strategies such as epsilon-greedy exploration (occasionally reviewing a transaction that the current policy would approve) or upper-confidence-bound (UCB) exploration, which prioritizes reviewing transactions the model is least certain about, allow the system to occasionally test uncertain cases rather than always relying purely on historical patterns. This exploration, combined with exploitation of proven detection rules, helps the system discover emerging fraud tactics earlier, continuously retrains the underlying model with fresh labelled examples, and ultimately improves long-term detection accuracy while reducing the system's vulnerability to novel fraud strategies that a purely static, rule-based engine would miss.

## Question 3

***Illustrate how Reinforcement Learning is used in a streaming platform for content recommendation. Discuss the impact of delayed rewards, user feedback, and changing preferences on learning performance.***

Streaming platforms use RL to personalize content recommendations by treating the recommendation engine as an agent that selects which titles to display to a user, observes how the user responds, and updates its policy accordingly. The environment is the user's ongoing viewing session and long-term relationship with the platform, and the state reflects their historical watch behaviour, genre and language preferences, time spent browsing before selecting a title, device context (mobile versus smart TV), and time of day.

## Actions and Rewards

Actions are the specific titles, thumbnails, or ordering of the recommendation carousel presented to the user, including which artwork variant of a title to show, since different thumbnails can strongly influence click-through rate. Rewards are derived from engagement signals such as clicks, watch time, completion rate, re-watches, and downstream subscription retention or cancellation, with higher rewards assigned to recommendations that keep the user engaged over a full viewing session rather than just generating an initial click.

## Impact of Delayed Rewards

A key challenge is that the true value of a recommendation is often not known immediately: a user may click a title but abandon it minutes later, or a single well-placed recommendation may influence subscription retention only weeks after viewing, while a series of poor recommendations may cause a user to quietly disengage over months. This delayed-reward problem makes credit assignment difficult — the agent must correctly attribute a long-term outcome, like renewed subscription, back to the specific earlier recommendation decisions that contributed to it. Techniques such as discounted cumulative reward, multi-step (n-step) returns, and eligibility traces are used so that the agent can properly propagate credit backward through a sequence of recommendation decisions rather than treating each recommendation in isolation.

## User Feedback and Changing Preferences

User feedback, both explicit (star ratings, thumbs up/down) and implicit (skips, replays, fast-forwarding, abandoning midway), continuously refines the state representation and reward signal. However, user preferences are non-stationary: tastes shift with mood, season, life events, or exposure to new genres, so an agent that only exploits historical preferences risks trapping the user in a narrow 'filter bubble' of similar content, which can reduce long-term engagement even if it maximizes short-term clicks. The agent must therefore keep exploring new content categories and occasionally test recommendations outside the user's established pattern, carefully balancing exploitation of known preferences with exploration of new genres, which is essential for maintaining recommendation quality, discovering shifting tastes early, and sustaining subscriber satisfaction over the long run.

# Question 4

***Evaluate the effectiveness of Reinforcement Learning in predictive maintenance compared to traditional rule-based approaches. Discuss risks and reliability concerns in industrial environments.***

Predictive maintenance aims to schedule equipment servicing before failure occurs, minimizing unplanned downtime and repair costs. Traditional rule-based approaches rely on fixed thresholds, for example servicing a machine every fixed number of operating hours, or triggering an alert whenever a sensor reading (vibration, temperature, pressure) crosses a static limit that was set once by an engineer based on historical averages. RL-based predictive maintenance instead learns an adaptive maintenance policy that decides, at each state of the equipment's condition — sensor readings, vibration signatures, temperature trends, wear indicators derived from vibration-spectrum analysis, and cumulative usage history — whether to continue operation, schedule maintenance at the next convenient window, or replace a component immediately.

## Effectiveness Compared to Rule-Based Systems

RL agents can capture complex, non-linear degradation patterns across many interacting sensor signals that fixed thresholds simply cannot represent, learning to balance the cost of unnecessary early maintenance (lost production time, spare-part cost) against the risk of costly unplanned failure (downtime, safety incidents, cascading damage to other components). Because the reward function penalizes both unplanned breakdowns and excessive/unnecessary maintenance interventions, the learned policy tends to be significantly more cost-efficient over time than static rules, which are typically set conservatively — leading to premature part replacement — or fail to generalize across different machines, operating environments, and load profiles, since a single fixed threshold cannot fit every unit in a heterogeneous fleet.

## Risks and Reliability Concerns

Despite these advantages, RL introduces risks that rule-based systems generally avoid. The agent's early exploration phase can involve trial-and-error decisions on real physical equipment, which is inherently risky in safety-critical industrial settings such as power plants, refineries, or heavy manufacturing lines, where an incorrect 'continue operation' decision could lead to catastrophic failure. RL policies can also behave unpredictably when the equipment enters operating conditions outside the range seen during training (distributional shift), and a poorly shaped reward function may lead the agent to under-prioritize rare but catastrophic failure modes simply because such events are infrequent in the training data. For this reason, industrial deployments typically combine RL with high-fidelity simulation-based pretraining (digital twins), hard safety constraints that override the learned policy in dangerous states, and continued human oversight and audit before an RL agent is allowed to make fully autonomous maintenance decisions.

## Question 5

***Analyze how Reinforcement Learning can be used to optimize transportation systems such as bus scheduling or route allocation. Define states, actions, and rewards, and discuss how real-time data improves performance.***

In public transportation, RL can optimize bus scheduling and route allocation by treating the transit control system as an agent that dynamically adjusts dispatch times, headways between buses, and route assignments in response to real passenger demand and traffic conditions, rather than following a rigid, pre-set timetable designed months in advance from average historical ridership.

## State Space

States include current passenger demand at each stop (from ticketing or camera-based counts), the location and occupancy level of every bus currently in service, prevailing traffic congestion on each route segment, time of day and day of week, weather, and historical ridership patterns for similar conditions.

## Action Space

Actions include dispatching an additional bus from the depot, holding a bus at a stop for a short period to prevent bunching, skipping a currently low-demand stop to save time, or reallocating a bus from an underused route to a route currently experiencing a demand surge.

## Reward Function

Rewards are designed around minimizing average passenger waiting time, reducing bus bunching (where two buses on the same route end up travelling close together, leaving long gaps elsewhere), maximizing on-time performance against the published schedule, and controlling total operating cost, with penalties applied for overcrowding, excessive idle time at stops, or buses running with very low occupancy.

## Impact of Real-Time Data

By continuously ingesting real-time GPS positions, ticket-tap ridership counts, and live traffic data, the RL agent can react to unexpected events such as accidents, sudden demand surges near a stadium or concert venue, or unplanned road closures far faster than a static timetable ever could. This real-time responsiveness improves overall service reliability, reduces average passenger wait times, and allows the transportation authority to make far more efficient use of a limited fleet of vehicles and drivers, ultimately improving both commuter satisfaction and the operational cost-effectiveness of the transit network as a whole.

## Question 6

***Apply Reinforcement Learning to dynamic pricing in e-commerce. Define the state space, action space, and reward function. Discuss challenges in balancing exploration and exploitation.***

Dynamic pricing in e-commerce uses RL to adjust product prices in near real time to maximize revenue or profit while remaining competitive against rival sellers and marketplaces. The pricing engine acts as the agent, and the marketplace — including competitor prices, customer demand elasticity, and inventory constraints — forms the environment the agent interacts with.

## State Space

The state includes current inventory levels for the product, competitor pricing gathered through market monitoring, recent demand trends and conversion rates, time-sensitive factors such as seasonality, festivals, or ongoing promotional campaigns, and customer segment characteristics such as browsing history or loyalty tier.

## Action Space

Actions correspond to setting a specific price point, or a discrete price adjustment such as increasing, decreasing, or holding the current price for a product at a given time interval; in more granular systems, actions may include personalized price offers shown to specific customer segments.

## Reward Function

The reward is typically defined as the realized revenue or profit margin from sales achieved at the chosen price over a fixed time window, with penalties applied for excess unsold inventory that must eventually be liquidated at a loss, or for setting a price so high that conversion rates and sales volume collapse.

## Exploration-Exploitation Challenge

A central challenge is that testing new, potentially higher-revenue prices (exploration) risks short-term losses if the price alienates price-sensitive customers or triggers a negative reaction, while always using known, historically profitable prices (exploitation) can leave revenue on the table and fail to adapt to shifting demand, changing competitor behaviour, or new market entrants. Techniques such as Thompson sampling, upper-confidence-bound bandits, or epsilon-greedy exploration are commonly used to test small price variations on limited customer segments before rolling changes out broadly, allowing the agent to continuously discover better pricing strategies while carefully limiting the financial risk and customer-experience cost of experimentation.

## Question 7

***Illustrate the use of Reinforcement Learning in smart city waste management systems. Identify the MDP components and discuss how continuous updates improve operational efficiency.***

Smart city waste management uses RL to optimize the collection and routing of waste-collection trucks based on real-time bin fill levels reported by IoT sensors, reducing unnecessary trips to empty or near-empty bins and improving overall service coverage compared to traditional fixed, calendar-based collection schedules that service every bin on the same day regardless of how full it actually is.

## MDP Components

The state space includes the fill levels of bins across the city as reported by ultrasonic or weight-based IoT sensors, the current locations of waste-collection trucks, real-time traffic conditions on candidate routes, and remaining time-window constraints for the day's collection shift. The action space includes deciding which bin to service next, which specific road segment or route to travel to reach it, or whether to deliberately skip a bin that is not yet sufficiently full to justify a stop. The reward function rewards efficient routes that prioritize collecting from full or near-full bins while minimizing total distance travelled and fuel consumption, and penalizes bins that are allowed to overflow (a public-health and cleanliness issue) as well as unnecessary stops at near-empty bins that waste time and fuel. The transition function captures how bin fill levels and truck positions evolve as actions are taken and as new waste continues to accumulate, and the policy maps the current, city-wide waste state to the single best next collection decision for each truck in the fleet.

## Continuous Updates and Efficiency

As sensor data streams in continuously throughout the day, the RL agent updates its policy to reflect real-time changes in waste-generation patterns, such as increased volume during festivals, weekends, or in specific commercial neighbourhoods, and reduced volume in residential areas during holidays. Over time, this continuous learning reduces the total number of collection trips required, lowers fuel and labour costs, prevents bin overflow and the associated public-hygiene complaints, and allows the waste-management authority to dynamically reallocate trucks to high-demand zones on short notice, meaningfully improving overall operational efficiency and service quality across the city.

## Question 8

***Evaluate how Reinforcement Learning can be used in network bandwidth allocation in telecommunications. Define states, actions, and rewards, and examine how the system adapts to dynamic conditions.***

In telecommunications, RL can dynamically allocate limited network bandwidth among competing users and applications, replacing static, manually configured allocation rules that cannot adapt quickly to rapidly changing traffic patterns across a cell tower, data centre link, or backbone network.

## State Space

The state includes current network load and congestion level on each link, the number of active users connected to a given cell or access point, the mix of application types being used (for example, latency-sensitive video calls versus best-effort messaging or background downloads), the Quality-of-Service (QoS) requirements associated with each active session, and the amount of available bandwidth across parallel channels or frequency bands.

## Action Space

Actions involve allocating a specific amount of bandwidth to a particular user or application, prioritizing certain latency-sensitive traffic types over best-effort traffic, or dynamically reassigning bandwidth from underused channels to currently congested ones, including switching a user between frequency bands.

## Reward Function

Rewards are structured to maximize overall network throughput and aggregate user satisfaction — measured through low latency, minimal buffering, and high perceived call/video quality — while penalizing congestion events, dropped connections, and unfair allocations that starve particular users or applications of the bandwidth they need to function acceptably.

## Adaptation to Dynamic Conditions

Because network demand fluctuates constantly, for example during peak evening hours or due to sudden spikes from a popular live-streaming event or breaking-news alert, an RL-based allocator continuously learns from real-time traffic metrics and adjusts its allocation policy accordingly, often on a sub-second timescale. This allows the network to maintain service quality under highly variable load far more effectively than fixed allocation schemes, improving overall resource utilization, reducing the frequency of congestion-related call drops or buffering events, and supporting a fairer and more efficient distribution of bandwidth as usage patterns continue to evolve with new applications and user behaviours.

## Question 9

***Analyze the application of Reinforcement Learning in portfolio management in finance. Discuss advantages over traditional models and challenges such as risk and delayed rewards.***

In portfolio management, RL treats the trading or asset-allocation strategy as an agent that decides how to distribute capital across a set of assets (equities, bonds, commodities, currencies) based on current market conditions, aiming to maximize long-term risk-adjusted returns rather than following fixed allocation rules derived once from traditional mean-variance (Markowitz) optimization, which assumes static statistical relationships between assets.

## States, Actions, and Rewards

The state reflects market indicators such as current asset prices, volatility measures, trading volume, macroeconomic signals (interest rates, inflation data), news sentiment, and the current composition and weighting of the portfolio. Actions correspond to buying, selling, or holding specific assets, or more generally rebalancing the portfolio weights across the available asset universe at each decision interval. The reward is generally based on realized portfolio returns, often adjusted for risk using measures such as the Sharpe ratio or Sortino ratio, with penalties applied for excessive volatility, large drawdowns, or transaction costs incurred from overly frequent trading.

## Advantages Over Traditional Models

Unlike traditional models that rely on static statistical assumptions about asset correlations, return distributions, and risk premia — assumptions that frequently break down during market stress — an RL agent can adapt its strategy as market regimes shift (for example, transitioning from a low-volatility bull market to a high-volatility bear market), learning complex, non-linear relationships between observed market signals and optimal trading actions directly from historical and live trading data rather than from a fixed mathematical model of the market.

## Challenges

Financial markets present significant challenges for RL: rewards are often substantially delayed, since the true value of a trading decision may only be realized weeks or months later once a position is closed; market data is extremely noisy, making it difficult to distinguish genuine, learnable patterns from pure randomness; and poor or overly aggressive exploration during training can expose the portfolio to significant real financial risk if tested directly on live capital. Overfitting to historical data is also a major concern in finance, since past market behaviour does not reliably guarantee future performance (a phenomenon well known as regime change), which requires careful risk management, extensive backtesting across multiple market cycles, and conservative position-sizing constraints before any RL-based strategy is deployed with real capital.

## Question 10

***Justify the use of Reinforcement Learning in renewable energy management systems. Define the RL framework components and discuss the impact of environmental uncertainty on decision-making.***

Renewable energy management, such as balancing solar and wind generation with grid demand and battery storage, is particularly well suited to RL because the availability of renewable energy is inherently variable and difficult to predict with simple rule-based control, making an adaptive, learning-based approach genuinely valuable for maintaining grid stability, minimizing cost, and maximizing the use of clean generation rather than costly fossil-fuel backup.

## RL Framework Components

The state space includes current energy generation output from solar panels and wind turbines, current battery storage charge level, real-time grid demand, short-term weather forecasts (cloud cover, wind speed), and prevailing electricity market prices. The action space includes decisions such as how much surplus energy to store in the battery, how much energy to draw from the main grid to cover a shortfall, or how much surplus generated energy to sell back to the grid at the current market price. The reward function rewards actions that minimize total energy costs, maximize the proportion of demand met from renewable generation, and maintain a stable, reliable power supply without interruption, while penalizing energy shortages that require emergency backup power, wasted surplus generation that could not be stored or sold in time, and unnecessary reliance on costly non-renewable backup sources such as diesel generators.

## Impact of Environmental Uncertainty

Because solar and wind output depend heavily on unpredictable and constantly changing weather conditions, the RL agent must continuously adapt its energy storage and distribution policy in response to updated generation forecasts and shifting demand patterns rather than relying on a single fixed operating schedule. This ability to learn from ongoing environmental uncertainty, instead of relying on rigid rule-based schedules designed for 'typical' conditions, is what justifies the use of RL in this domain: it enables more efficient use of available renewable resources, meaningfully reduces reliance on fossil-fuel backup generation and the associated cost and emissions, and improves the overall reliability and cost-effectiveness of the energy system as conditions fluctuate hour to hour, day to day, and season to season.